Candidate Number

Candidate Name____________________________

INTERNATIONAL ENGLISH LANGUAGE TESTING SYSTEM

Academic Reading

Additional materials:

Answer sheet for Listening and Reading
Time 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this question paper until you are told to do so.
Write your name and candidate number in the spaces at the top of this page. Read the instructions for each part of the paper carefully.
Answer all the questions.
Write your answers on the answer sheet. Use a pencil.
You must complete the answer sheet within the time limit.
At the end of the test, hand in both this question paper and your answer sheet.

INFORMATION FOR CANDIDATES

There are **40** questions on this question paper.
Each question carries one mark.

READING PASSAGE 1

*You should spend about 20 minutes on Questions **1–13**, which are based on Reading Passage 1 below.*

The history of colours and the meanings people have given them

The use of colours has a long history in human development. Prehistoric paintings in caves were found to contain yellow and red earth, white chalk, and a black made from the soot of burned animal fat. Over time, new colours were invented, using a wide range of materials and processes; for example, Chinese yellow was made from the resin of the gamboge tree, and saffron yellow in India from a flower, the Crocus sativus. Archaeological evidence tells us that these societies did not operate in the production of these yellows. In Europe in the Middle Ages (470–1500 AD), red was made from kermes insects, but when the Spanish explorers returned from the Americas in the 15th century, they brought not only gold, silver, and new bookshelves, but also a new shade of red. What was different about this new shade was that it was the result of a process involving a species of beetle.

Such colours were expensive. In ancient times, more than 10,000 murex shellfish had to be crushed to make a single gram of purple. As a result, purple became the colour of kings and emperors in ancient Persia and Rome, and was also worn by priests on ceremonial occasions. In Europe, during the Renaissance (1300–1600 AD), patrons would state in contracts that painters were required to use expensive pigments to add value to the works they commissioned. For instance, they might ask that the artist use gold paint or ultramarine, which was made from the semi-precious gem called lapis lazuli. This had to be imported from 'beyond the sea', which is what 'ultramarine' means in Latin.

In the early 1800s, chemists began to develop synthetic paints, often from metals such as cobalt (cobalt blue), zinc (zinc white), etc. Without these new colours, Impressionism and other aspects of modern art development would not have been possible. By the mid-1800s, another development was the use of portable tubes to contain the paint and keep it fresh. Today, still more pigments are being invented, many of synthetic organic origin, adding new colours, greater transparency for mixing or glazing, greater lightfastness, and so on. Colour printing techniques, colour film, and video also continue to improve, and modern software is, of course, the newest colour resource. Photo editing software makes the colour manipulation of photographic images widely available. Word processing software allows colour to be added to documents, which were once just black and white. The number of colours companies can produce has risen from 16 shades of grey to millions of colours – more than anyone could possibly need.

There are many different meanings that people have given to colours. This is a process that has gone on for much of human history and has not yet come to an end. Green being associated with environmental protection is a good example of this. In Europe, the colour black has been used to express a variety of meanings, for example, formality and elegance, among many others. However, black is not the only colour that represents death. In China and some other Asian countries, the colour of sickness and death is white, whereas it is red which is traditionally used for weddings. The same colour represents danger in many European countries. There is always a reason why a given colour becomes associated with certain feelings or ideas, but that does not mean the association is universal.

Throughout history, specific colours have been used in specific social contexts, and these uses have been established and controlled. A clear example of this can be seen in the products developed and marketed for children. Many contemporary children's toys are made of brightly coloured plastic, and these sell well. Books for very young children also tend to feature bright primary colours (red, yellow, and blue) because young children find these appealing. The reason for this attraction was explained by colour psychologists. Jonas Cohn, for instance, writing in the 1890s, said that preference for strong, pure colours is a basic human instinct, and therefore this preference is strongly present in children.

Educators, designers, and manufacturers adopted Cohn's idea as part of developing a new culture of childhood with its own dress code, its own literature – and its own colour schemes. Lately, ideas have altered, and children are introduced to adult culture much earlier. Consequently, bright reds, yellows, and blues have to some degree been replaced by mixed colours such as pinks, mauves, and oranges.

Questions 1-6

Do the following statements agree with the information given in Reading Passage 1? In boxes ***1-6*** *on your answer sheet, write:*

TRUE *if the statement agrees with the information*
FALSE *if the statement contradicts the information*
NOT GIVEN *if there is no information on this*

1. The shade of red brought back by 15th-century explorers was more popular with the public than the red made from kermes insects.

2. During the Renaissance, some European artists were obliged to use low cost materials.

3. The colours invented by chemists in the 1800s led to major new art movements.

4. People today continue to give new meanings to colours.

5. In Europe, there are an equal number of positive meanings of black as there are negative ones.

6. Jonas Cohn believed children naturally like dark colours.

Questions 7-13

Complete the notes below.
Choose ***ONE WORD ONLY*** *from the passage for each answer.*
Write your answers in boxes 7-13 on your answer sheet.

Making and Using Colours

Making paints

• The earliest paints discovered in **7 ....** were made from animal products, chalk or different soils.

• In the Middle Ages, red was made from either kermes insects or a kind of **8 ....**

• Purple was:

made from shellfish

expensive

used mostly by rulers and **9 ....**

• In the Renaissance, ultramarine was made out of a type of **10 ....**

• In the 1800s, chemists started to make colours by using different **11 ....**

• In the mid 1800s, colours were put into **12 ....** which made them easy to transport.

Colours and different countries

• Red is worn at **13 ....** in some countries but is also used to warn people of danger.

READING PASSAGE 2

You should spend about 20 minutes on Question 14–26, which are based on Reading Passage 2 below.

Questions 14–20

*Reading Passage 2 has seven sections, **A–G**.*
Choose the correct heading for each section from the list of headings below.

List of Headings

i Packaging that allows consumers to use more of the product

ii The many reasons for changes in packaging

iii Packaging that includes utensils like spoons and forks

iv Packaging that can be consumed with its contents

v Packaging that doesn't take up a lot of space

vi The history of food packaging

vii Packaging that indicates when food is beginning to go rotten

viii Packaging that can be reused

ix Using packaging to communicate ideas and attract customers

x Packaging that vanishes or decomposes quickly

14. Section **A**
15. Section **B**
16. Section **C**
17. Section **D**
18. Section **E**
19. Section **F**
20. Section **G**

The Future of Food Packaging

A Food companies spend a lot of time thinking about packaging as a way to represent their brand and make consumers love their products. More than cosmetic details like labels or logos, a food's container can suggest playfulness, environmental consciousness, or high technology. Packaging provides consumers' first and last impressions of a product.

B Fortunately for consumers, advances in food packaging technology promise to make food storage and preparation simpler. Consumers are growing ever more obsessed with sleek technology, reducing food waste, and minimizing health risks to themselves and the environment, and in response, food packaging is undergoing some interesting developments. Food-packaging designers are hard at work to make groceries environmentally friendly, compact, and functional.

C Edible packaging has been proposed as one solution to consumers' environmental concerns because it leaves no waste. The fried chicken chain KFC, for example, is experimenting with an edible coffee-cup. Made from a hard cookie lined with heat resistant white chocolate, the cup is KFC's attempt to reduce its landfill waste and satisfy customers who are concerned that businesses produce products that support food-system sustainability and innovation. Another company, Stonyfield Organic, introduced frozen yogurt 'pearls' using edible food wrappers. The skin has no flavor and can be rinsed and eaten with the yogurt (much like a grape), eliminating the need for plastic spoons or wrappers for those on the go. It seems that people still want a layer of sanitary protection, though, because the spherical yogurts are still sold in containers. Similarly, a company called Loliware has sold gelatin-based edible drinking cups for two years, in flavors like Madagascar vanilla and tart cherry. And last year, the company introduced edible cupcake wrappers made from potato starch. Even with these new advances, one decade-old edible invention continues to see great success — the clear paper used to wrap Japanese Bolan Rice Candy.

D Another development has been nonstick packaging, which can eliminate the 3 to 15 percent of sauces left inside traditional bottles, like those last trails of ketchup and mayonnaise that won't come out no matter how hard you shake or squeeze. Liquiglide — which refers to both a company and the product it makes — is a permanently wet slippery surface that allows viscous matter to slide off easily. Liquiglide estimates that its coating could prevent 1 million tons of food waste each year. The coating is made from components approved by the US government, and, according to the company, does not affect the taste or smell of food. Another benefit of Liquiglide is that it prevents carbonated beverages from going flat.

E New safety-enhancing packaging is now addressing the issue of food spoilage. It might seem this was solved long ago with the advent of the 'Sell-By' date stamp, which was meant to warn customers of when the food would no longer be considered suitable for consumption. But after a Harvard report revealed that sell-by dates are often inaccurate because of variances in transportation and grocery store policies, and that they also contribute to America's food-waste problem, the stamps have lost credibility in the food industry. In response, scientists are working on food packaging — like plastic wrap — that is 'active and intelligent' with tiny embedded sensors to first detect and later prevent food from spoiling. This advancement in food packaging draws upon nanotechnology, which is dedicated to making electronic sensors tiny and inexpensive. Of course, one could also rely on what has saved humankind from food poisoning for millennia — the senses. If it smells bad, throw it away.

F Another emerging technology is packaging that disappears. MonoSol, LLC creates food-grade water-soluble bags for pre-portioned oatmeal or rice. When boiled, the bag dissolves, leaving only the food, ready to eat. MonoSol, LLC is marketing this to professional kitchens as well as consumers, as a way to pre-measure meal ingredients and ensure consistency regardless of who is cooking. Tomorrow Machine, a Swedish design firm, has plans for three disappearing packages: an oil bottle made from hardened sugar that cracks open like an egg; a fruit juice container made from seaweed that 'will wither at the same rate as its content'; and a peelable beeswax package for rice or other dry goods.

G Super-compressed packaging serves the increasing number of city dwellers who generally have tiny kitchens and no storage room. Additionally, many people living in urban areas walk home from grocery stores, and they need lighter and smaller packages. Tomorrow Machine is also working on a compressed tiny package that expands into a full-size bowl when hot water is added.

Questions 21–23

*Look at the following companies (Questions 21–23) and the list of packaging developments below. Match each company with the correct packaging development, **A–F**.*
*Write the correct letter, **A–F**, in boxes 21–23 on your answer sheet.*

List of Packaging Developments

A. a container that gets bigger under certain conditions

B. packaging that vanishes in hot water

C. a material that prevents the waste of sticky condiments

D. packaging made from a vegetable product

E. a sweet container for hot drinks

F. round, flavourless containers that can be consumed

21. KFC
22. Loliware
23. Tomorrow Machine

Questions 24–26

Complete the sentences below.
*Choose **ONE WORD ONLY** from the passage for each answer.*
Write your answers in boxes 24–26 on your answer sheet.

24. In Japan, ________ has been used as edible packaging for years.

25. Liquiglide reduces food waste, especially in bottles containing various _____.

26. Tomorrow Machine is planning to use solid ________ as a container for oil.

READING PASSAGE 3

You should spend about 20 minutes on Questions 27–40, which are based on Reading Passage 3 on pages 7 and 8.

All in the family

A person's brothers and sisters can have a powerful ef ect on their adult behaviour

We can choose our friends, the saying goes, but we cannot choose our family of origin. Our parents provide the genetic material and make powerful early role models, but even more influential in determining what kind of adult we will become are our brothers and sisters — our siblings. They occupy a position of unique intimacy in our lives and cast a longer shadow than many recognise or are prepared to acknowledge. By turns, enraging and lovable, familiar and mysterious, our brothers and sisters are the human beings who people our first social relationships. And a sibling relationship is the most enduring relationship many of us will ever have — no less emotionally intense than the bonds we form with our spouses and our own children — if only because they started when we were so young.

Surprising new research by Dr Gene Brody found that having older children who do well in school and are well liked by other children leads to parental 'basking' — increasing mothers' self-esteem. It is clear that in turn this is associated with more positive parenting of younger children, who display fewer behavioural problems as a result. Conversely, parents who get a difficult first child may in turn experience a negative spiral of household tension.

International sports star Piri Weepu is a rugby hard man, but at least some of his tough temperament was forged long ago in the family home. His older brother Billy started rugby — tackling little Piri when he was just four — to 'harden him up'. It seemed to work: Piri followed his brother in playing for the under-sevens league before he even started school. There are as many such stories as there are families: in the formative hothouse of the family, siblings interact in complex ways with the powerful forcefields of parents, genetics and personality already at work. It starts with birth order, which determines play roles: the firstborn leader, the middle-child mediator and the rebellious youngest. These are roles that can stick for life.

Clinical psychologist Claire Cartwright says she has clients who believe, and it might not have been true, that they were treated differently from their siblings, that one child in the family was preferred over others. They are quite convinced that the parental judgements on them all those years ago were harsher, and that they were the ones who always got into trouble. Later, in the workplace, such a person might be particularly defensive, and so behave in a way more prone to attracting harsh judgements.
However, the research shows that being Mummy or Daddy's favourite has its own traps. If you were the easy-going, co-operative child at home, you probably excelled at pleasing parents and, later, bosses. But you might, as an adult, find it hard to be assertive with authority figures and lack direction. On the other hand, the sibling who is disruptive and annoying as a child may be able to reinterpret that role later, turning those attention-seeking characteristics into strengths like determination and leadership. How is it that full siblings, despite sharing DNA, can turn out so differently? One answer is that, in fact, each sibling grows up in a different family, a unique micro-culture. For example, the firstborn is, for a while, an only child, and therefore has a completely different experience of the parents than those born later. The parents themselves are growing up too, weathering hardship or good fortune, so one sibling might experience stability and closeness while another might be raised in the midst of crisis.

Of course, there are many positive aspects of sibling relationships. Annette Henderson, a lecturer in psychology, says firstborn children learn vocabulary more quickly than their siblings because they are not competing to spend one-on-one time with parents. But younger children in turn benefit from unintentional instruction from their bigger brothers and sisters, acquiring entire phrases and an understanding of social concepts such as politeness. Similarly, a Cambridge University study of 140 children found that even when there is a great deal of conflict, siblings will still create a rich world of play and make-believe that extends them developmentally. Love-hate relationships were common among the children, but even those who fought the most had as many positive interactions as the other sibling pairs. It is also true that children compete for parental attention by making themselves different from their brothers and sisters, particularly if they are close in age. A 2003 research paper studied adolescents from 185 families over two years, finding that those who changed, to differentiate themselves from their siblings, increased the amount of warmth they gained from parents, and also developed stronger personalities and a better sense of their own identity.

Another consideration is that many families today are smaller and more intimate than their historical counterparts. This may be advantageous because siblings tend to know each other better and remain lifelong friends compared to children from bigger or more widely spaced families, who perhaps lose touch with one another after leaving home. Then there was a 2010 American study which found that having a sister, whether younger or older, meant that 10- to 14-year-olds were less likely to feel lonely, unloved, self-conscious or fearful. The same study found that having a loving sibling of either gender promoted good deeds, such as helping a neighbour around the house or helping other children at school.

Questions 27-30

*Do the following statements agree with the claims of the writer in Reading Passage 3? In boxes **27-30** on your answer sheet, write:*

YES *if the statement agrees with the claims of the writer*
NO *if the statement contradicts the claims of the writer*
NOT GIVEN *if it is impossible to say what the writer thinks about this*

27. Parents and siblings play equal roles in shaping people's adult personalities.

28. People's first social relationships tend to be with siblings of their own gender.

29. People's relationships with their siblings tend to receive less attention when they marry.

30. Parental 'basking' is advantageous for later siblings.

Questions 31-34

*Choose the correct letter **A**, **B**, **C** or **D***
Write the correct letter in boxes 31-34 on your answer sheet.

31 What point does the writer make about birth order in the third paragraph?

A. Its role is underestimated by parents.
B. It may shape the characteristics of adults.
C. Its importance decreases as children grow up.
D. It has more influence on personality than genetics.

32 What are we told about some of Claire Cartwright's clients?

A. Their parents treated them fairly.
B. They were treated harshly by their siblings.
C. Their perceptions of childhood may be inaccurate.
D. They have closer relationships with their siblings as adults.

33 What is the writer's main point about siblings in the fifth paragraph?

A. Parents tend to repeat their own parents' mistakes.
B. Favoured children are likely to make good leaders as adults.
C. Negative characteristics in children can become positive in adults.
D. Attention-seeking children are unlikely to please bosses in later life.

34 Which of the following best summarises the writer's argument in the sixth paragraph?

A. Many children have a favourite parent.
B. Parenting skills improve with later children.
C. Siblings are born with different personalities.
D. The family environment may change over time.

Questions 35-40

*Complete the summary using the list of phrases, **A-K**, below.*
*Write the correct letter, **A-K**, in boxes 35-40 on your answer sheet.*

The positive aspects of sibling relationships

Annette Henderson suggests that young children's interactions with older siblings involve **35** .... Also, a Cambridge study found that siblings still enjoy **36**.............. even when they have a tendency to fight. Research conducted in 2003 suggests that children develop **37** .... as a result of seeking attention. Another factor is the trend for smaller families, which may mean today's siblings enjoy **38** .... as adults. A 2010 American study linked having a sister with **39** .... among children aged 10-14. Finally, having a sibling may promote **40** .... according to the same study.

A. parental love	**B.** imaginative games	**C.** physical development
D. generous behaviour	**E.** strong marriages	**F.** greater individuality
G. learning experiences	**H.** career success	**I.** closer relationships
J. healthy competition	**K.** mental well-being	